

Week 6 – Machine Learning Model Architecture Design (20260618)

Week 7 - Advanced Analysis & Tool Development

Report date: 20260618 | **Pipeline version**: v1.0

1. Tool Configuration

- Model artifact: week6_approach2_xgboost_v1.0.joblib
- Reference date: 2024-12-31
- Input feature count: 23
- Baseline model price: 16.098978
- Closed-form reference: 33.429823
- Monte Carlo reference: 33.602667

2. Extreme Scenario Tests

scenario	model_price	delta	delta_pct	closed_form_quote	mc_quote
baseline	16.098978	0	0	33.429823	33.602667
volatility_spike_50pct	16.734989	0.636011	0.039506	33.429823	33.602667
rate_hike_2pct	16.098978	0	0	33.429823	33.602667
sentiment_shock	16.081207	-0.017771	-0.001104	33.429823	33.602667
combined_stress	17.38422	1.285242	0.079834	33.429823	33.602667

3. Sensitivity Grid

feature	value	model_price	delta	delta_pct
vix	12.145	15.147018	-0.95196	-0.059132
vix	15.615	15.113082	-0.985896	-0.06124
vix	17.35	16.098978	0	0
vix	19.085	16.098978	0	0
vix	26.025	17.38422	1.285242	0.079834
sentiment_7d	0	16.098978	0	0
sentiment_7d	0.005333	16.098978	0	0
sentiment_7d	0.105333	16.098978	0	0
sentiment_7d	0.205333	16.098978	0	0
sentiment_7d	0.355333	16.098978	0	0
sentiment_20d	0	16.002396	-0.096582	-0.005999
sentiment_20d	0.04583	16.002396	-0.096582	-0.005999

feature	value	model_price	delta	delta_pct
sentiment_20d	0.14583	16.098978	0	0
sentiment_20d	0.24583	16.098978	0	0
sentiment_20d	0.34583	16.098978	0	0
r	0.0358	17.633144	1.534166	0.095296
r	0.0458	16.098978	0	0
r	0.0558	16.098978	0	0
r	0.0658	16.098978	0	0
q	0.01419	16.098978	0	0
q	0.01919	16.098978	0	0
q	0.02419	16.098978	0	0
T2	0.3	16.098978	0	0
T2	0.4	16.098978	0	0
T2	0.5	16.098978	0	0
7 more rows omitted				

4. SHAP Impact Summary

feature	mean_abs_shap
vix_ma_ratio	0.696841
vix	0.46509
S	0.101307
q	0.064257
vix_jpm_corr_20d	0.033735
vol_ratio_20_60	0.025271
news_count_7d	0.023548
return_5d	0.002082
sentiment_20d	0.00005
sentiment_7d	0

5. Observations

- The pricing tool now exposes a direct quote, a closed-form reference, and a Monte Carlo reference for the same contract.
- Scenario tests include a 50% volatility spike, a 2% rate hike, a sentiment shock, and a combined stress case.
- Sensitivity grids are centered on the latest usable market row so the tool can be refreshed with new daily data.